

POORANI RAVINDHIRAN

poorani@umich.edu | [+1-734-881-3641](tel:+17348813641) | [Linkedin](#) | [Website](#)

EDUCATION

University of Michigan, Ann Arbor, Michigan

Master of Science in Robotics

May 2020

GPA: 3.63/4.00

Relevant courses: Machine Vision, Deep Learning, Linear Algebra, Probability, Calculus, Multi-view Geometry, Self-driving Cars

SSN College of Engineering, India

May 2018

Bachelor of Engineering in Electronics and Communication Engineering

GPA: 9.04/10.00

Publication: [Automated indoor waste management system employing a mobile robot](#) in IEEE International Tech Conference

EXPERIENCE

Software Engineer at Qualcomm, San Diego, CA

Apr 2021- Aug 2023

- Created a TensorFlow-based deep learning pipeline on Google Cloud to detect and classify video artifacts with a 93% accuracy. Engineered a training dataset of 50k images, using image manipulation techniques to simulate visual artifacts.
- Enhanced the efficiency of manual video quality validations by 80% through contributions to scaling efforts. Collaborated on two additional deep learning projects focused on detecting audio-video sync and frame order.
- Developed Python and C++ automation tools for video quality validation in the encoding and decoding pipelines.

Robotics Engineer at Fuji Innovation Lab, Seattle, WA

Oct- Dec 2020

- Simulated a warehouse robot in Gazebo; Implemented gyro-odometry and geo-fencing with sensor fusion analysis.
- Achieved precise instance segmentation and visual tracking of inclined surfaces from 3D point clouds using a clustering and region growing algorithm with RANSAC, ensuring an accuracy of 0.02 meters in ramp width and of 10° in slope angle.
- Validated pose estimation and loop closure using RViz and evaluated impact of static and dynamic objects on G-Mapping.

Autonomous Vehicle Testing Intern at Qualcomm Research Center, San Diego, CA

May- Aug 2019

- Analyzed sensor fusion module performance under changing camera tracker's frequency, map density and ego pose error.
- Streamlined 180° sensing and identified two redundant cameras, reducing costs and computational complexity by 20%.

SOFTWARE SKILLS

Languages: C++, Python, Julia, MATLAB, embedded C, SQL

Libraries: PyTorch, TensorFlow, Keras, OpenCV, Caffe, PCL

Platforms: Google Cloud AI, AWS, Linux, Docker, CUDA

Frameworks: Git, ROS, Gazebo, JIRA, Jenkins, P4V

FEATURED PROJECTS

Anti-aliasing NeRF for novel view synthesis of 3D scenes with adaptive rendering

- Implemented vanilla [NeRF](#) from scratch using Pytorch on a Nvidia A100 GPU, yielding a 22.28 PSNR and a 0.92 SSIM on synthetic Lego data. Enhanced memory efficiency via lazy data loading and optimized ray batch processing with parallelism.
- Added adaptive rendering module via selective sampling through image segmentation using [FastSAM](#) on [TanksandTemples](#) data, resulting in finer rendering in areas of interest with a 6% decrease in training time.
- Enabled scale-aware learning by incorporating cone tracing and integrated positional encoding from [MipNeRF](#), reducing model parameters by 47% and improving test SSIM by 15% on 1/8th resolution images while eliminating aliasing artifacts.

Contextually guided few-shot semantic segmentation

- Implemented a generalized 5-shot segmentation pipeline via a 3-step meta learning method in Pytorch, achieving a 5% improvement in mIoU score for both novel and base classes in [Pascal VOC](#) data.
- This involved implementing the [Pyramid Scene Parsing](#) model with a global scene-level prior to reduce misclassification. enhanced it with limited novel class support samples to learn their contextual relationship, and dynamically extending this learning to query samples to mitigate bias from limited support data and generalize segmentation results.

Deep loop closure in ORB-SLAM2 framework for scene understanding

- Built a denoising autoencoder network in Keras trained on [Places2](#) dataset to generate HOG like descriptors ([dHOG](#)).
- Modified the loop closing module in ORB-SLAM2 to use the generated dHOG features instead of traditional ORB features with BoW representation, resulting in a 30% faster query time and improved precision-recall curve.
- Evaluated quantitatively on CampusLoop data and qualitatively with a 2D simulation on [KITTI](#) dataset in RViz.

Anomaly detection using a neural rendering model with tensor layers

- Implemented a CNN-guided [generative model](#) for artifact detection in the [RoboPol](#) astronomical dataset, effectively identifying out-of-distribution samples (even with lower variance) through joint latent variable likelihood learning.
- Overcame training constraints by replacing fully connected layers with tensor layers, preserving the data structure, resulting in a better detection accuracy with a 53% reduction in training time and 76% fewer trainable parameters.